

# PDF ANONIMIZADO

ID publico: case\_0194

Paginas del documento: 4

Copia anonimizada para verificacion metodologica.

Se removieron portada editorial, titulo, autores, afiliaciones, correos, DOI, URLs y metadatos del PDF original.

Las paginas restantes conservan el contenido util para revisar metodos, resultados, tablas y conclusiones.

---

Nota: la anonimización automatica prioriza reducir la identificacion directa del articulo. Los casos de una sola pagina o diseno no convencional deben revisarse manualmente antes de una publicacion abierta.

technique can replace wrist arthrographer magnetic fields, MRI has proven to for diagnosing peripheral tears, with

However, some authors consider wrist AV to be the gold standard for diagnosing TFCC injuries to assess intrinsic wrist ligament injuries, as it allows direct visualization of the wrist joint. Although considered an invasive method, diagnostic arthroscopy is indicated for patients with unequivocal or negative imaging tests, but with a history, symptoms, and clinical examination consistent with the disease.

Considering the different treatment methods, initially conservative treatment with immobilization, modification of activities, corticosteroid injections, and the use of painkillers for a few weeks should be considered first. When symptoms do not go into remission, surgical procedures are recommended. Currently, there is no standard gold standard procedure, and no surgical procedure has been found to be superior.

PRP injection is a recent and commonly used option in the treatment of acute and chronic tendinopathies because it promotes cell viability and collagen synthesis by providing the growth factors TGF- $\beta$ 1 and PDGF-AB.

The *outside-in*, *inside-out*, or *all-inside* arthroscopic methods allow for appropriate treatment when indicated, where repair with an anchor to perform transosseous suturing is the most commonly used procedure. This feature has the advantage of avoiding open capsulotomy with capsular flap of the original method. This instrument minimizes scarring, soft tissue fibrosis, and joint stiffness. Considering the evaluation of the results, good and excellent results are generally obtained in 60% to 90% of cases.

To evaluate the results, we found that the Visual Analog Scale (VAS), modified Mayo score, arm, shoulder, and hand disability scores, grip strength, and range of motion (ROM) are the primary assessment tools used in the treatment of patients with TFCC injuries.

Therefore, this study aims to evaluate the clinical outcome of arthroscopic surgical treatment of conditions affecting the triangular fibrocartilage of the wrist. As a secondary objective, the authors will evaluate the correlation between the findings of the MRI scans and those found during wrist VA.

## MATERIALS AND METHODS

Initially, this research project was submitted to bioethical evaluation by the Research Ethics Committee of Plataforma Brasil and approved for implementation by CAEE opinion 81673924.2.0000.0087. This is a retrospective cross-sectional cohort study whose sample included patients with TFCC lesions treated surgically between February 2021 and September 2023.

The following criteria were established for inclusion of patients in this study.

1. Patients of both sexes.
2. Aged between 18 and 60 years old.
3. Minimum follow-up period of 6 months.
4. With a diagnosis of traumatic (partial or extensive) or degenerative injury.
5. No previous trauma such as styloid process fractures of the ulna, distal radius fractures, or ARUD injury.
6. No previous history of inflammatory disease
7. Patients with failure of conservative treatment.

Therefore, our case series consisted of 28 patients, 21 (75.00%) men and 7 (25.00%) women, with a mean age at the time of surgery of 39.60 years (minimum of 20 years and maximum of 58 years) and a mean follow-up time of 13.6 months (minimum of 6 months and maximum of 20 months). Twelve (42.86%) were performed on the right wrist and 16 (57.14%) on the left wrist.

Arthroscopic wrist surgeries at the research center in question are performed under general anesthesia with a tourniquet on the upper limb being operated on, with a pressure of approximately 250 mmHg. After sterile preparation, the patient's hand is placed in a traction device and a force of approximately 6 to 7 kg is applied. Using the dorsal approach, portals 3–4 and 4–5 are used for arthroscopy and *shaving*. A *shaver* is used to release the TFCC from the capsule in the *foveal* region. Peripheral and deep tears are visualized through portals 3–4 after release of the TFCC, and instability of this structure is assessed with the aid of a hook. The drill guide is passed through the ulnar surface toward the fovea of the ulna, then a guide wire is passed through and a 2.5 mm suture anchor is inserted. The grip is then released and the repair between the TFCC and the fovea is secured with three knots. The knots are positioned inside the sheath of the ulnar extensor tendon of the carpus to prevent skin irritation. When ligament reconstruction is not necessary, only debridement is performed. After removal of the arthroscope, the portals are sutured with 5-0 nylon. Finally, a sterile dressing is applied and the pneumatic tourniquet is deflated. During outpatient follow-up, the stitches are removed after two weeks and the axillary and palmar immobilization is removed after two weeks when ligament reconstruction is not necessary, or after three weeks when it is necessary. After this period, physical therapy treatment begins.

The Quick-Dash functional assessment method consisting of 11 questions was used. With 11 questions, the tool measures the impact of injuries in these regions, assessing functional limitations, pain, and the ability to perform daily activities.<sup>3</sup> Its main function is to monitor the progress of treatment and compare the effectiveness of interventions, and it already has a validated Brazilian version.<sup>4</sup> Responses to QuickDASH were presented based on the relative frequency of responses. The number of injuries identified, as well as their origins and characteristics, in addition to the need for sutures, were presented according to their absolute and relative frequencies in relation to the total sample size.

## RESULTS

In the collective evaluation of responses to Quick-Dash, when patients were asked if they could open a new jar or one with a very tight lid, 76.7% responded that they had no difficulty, while 23.3% responded that they had little difficulty. Regarding the ability to perform heavy household tasks, such as washing walls or floors, 63.3% reported no difficulties, 26.7% reported little difficulty, while 10.0% reported moderate or severe difficulty. Regarding the ability to carry a bag or briefcase, 93.3% reported no difficulties, while 6.7% reported little or moderate difficulty. When asked about their ability to wash their own backs, 86.7% reported no difficulty, with 13.3% reporting minor difficulty. When asked about their ability to use a knife to cut food, 93.3% of patients reported no difficulty, with 6.7% reporting minor difficulty. Regarding the ability to participate in recreational activities that require some strength or impact on the arms, shoulders, or hands, such as playing volleyball or hammering,

erately affected. When asked whether their work or even limited their regular k, 76.7% said there were no limitations, itations. Pain in the arm, shoulder, or hand was not reported by 73.3% of patients, while mild pain was reported by 26.7%. Skin discomfort (pinching) in the arm, shoulder, or hand was not reported by 90% of patients, being present in 10% of the sample (mild discomfort). Finally, when patients were asked whether they had difficulty sleeping because of pain in their arm, shoulder, or hand, 90% reported no difficulties, while 10% reported minor difficulties.

Soon after administering the Quick-Dash questionnaire, we obtained 18 (64.28%) excellent, 7 (25.00%) good, and 3 (10.72%) satisfactory results. MRI identified 4 (14.29%) extensive lesions and 24 (85.71%) partial lesions, while VA identified 2 (7.14%) extensive lesions and 26 (92.86%) partial lesions. With regard to agreement between RM and VA, we observed 26 (92.86%) cases.

## DISCUSSION

Suspected TFCC damage is diagnosed through appropriate physical examination, which allows for a reliable diagnosis of foveal tears. Palpation of the bone near the ulnar styloid often causes pain, and when combined with a positive stress test, such as foveal compression, indicates a rupture. Occasional additional clicks may also indicate the presence of ARUD instability.

However, it is indisputable that diagnosis depends on resources such as radiography, arthrography, MRI, magnetic resonance arthrography (MRA), and wrist VA. However, there is no consensus regarding the sensitivity, specificity, and accuracy of such diagnostic imaging resources, despite their frequent use.

The diagnostic values of this modality range from 71 to 100% for central and radial ruptures of the TFCC. However, the diagnostic performance of conventional MRI for detecting peripheral tears is only 17%. This percentage is justified by the presence of high-sign vascularized fibrous tissue between the two ulnar insertions mimicking the rupture.

According to some studies, MRI is up to 95% accurate in detecting foveal tears of the TFCC complex.<sup>5</sup> This technique is sensitive for diagnosing such lesions, but according to some authors, arthroscopy appears to be the most accurate method for diagnosis.

However, according to hand orthopedic surgeons, the gold standard for evaluating intrinsic wrist ligament injuries is arthroscopy. His argument relates to the possibility of directly visualizing the articulation of the wrist. However, we also found divergent opinions questioning the degree of diagnostic confidence. It is considered an invasive and expensive method, so VA would be restricted to patients with unequivocal or negative imaging studies despite having a history, symptoms, and clinical examination consistent with the disease.

For ARM, we found the study by Haims et al.<sup>6</sup>, which evaluated 86 MRI exams and found sensitivity of 17%, specificity of 79%, and accuracy of 64%. In the study by Morley et al.<sup>7</sup>, we found 54 MRI studies that demonstrated sensitivity of only 44% and specificity of 87%.

The diagnostic accuracy of MRI increases with magnetic field strength (3 Tesla), coil technology, and pulse sequencing compared to examinations on 1.5 T machines.

In a study conducted by Smith et al.<sup>5</sup> in a systematic review that

can cause pain and discomfort in the wrist due to the administration of intra-articular contrast. Due to synovial irritation. In rare cases, serious complications such as joint infection should be remembered. We believe that the absence of a standardized protocol for performing MRI in the evaluation of TFCC may contribute to low diagnostic accuracy. Variations in MRI parameters, such as magnetic field strength and contrast agent use, are not effective in detecting TFCC lesions. On the other hand, VA performed with specific portals allows a comprehensive assessment of the wrist structures, facilitating accurate classification of injuries according to Palmer's classification.<sup>9</sup> In this case, standardization of MRI could contribute to improving diagnostic accuracy and allow replication of results in different centers.

The importance of wrist AV in the diagnosis and treatment of TFCC injuries is indisputable. We highlight its importance in allowing arthroscopic inspection to achieve a correct diagnosis of injuries, as well as their morphological characteristics such as location, shape, and dimensions. Such assessments are possible thanks to the magnified view of the structures studied and the possibility of using various portals. They also allow visualization of the radiocarpal and midcarpal joints to diagnose and treat coexisting injuries.

Furthermore, arthroscopic techniques theoretically offer advantages, including smaller incisions, less tissue dissection, faster recovery, and greater patient satisfaction.

Our study compared the diagnostic efficacy of MRI in relation to VA. We observed that MRI detected TFCCP lesions in 100% of cases, with 4 (14.28%) being extensive and 24 (85.71%) partial. The VA showed 2 (7.14%) extensive lesions and 26 (92.85%) extensive lesions. However, the degree of agreement between the tests used in our study was 92.85%, with two cases showing no agreement regarding the extent of the damage. We agree with the opinion of several authors that there are advantages to VA. Still, the degree of agreement between the resources evaluated was considered excellent according to the Kappa classification (with the highest number of categories). The population sample evaluated in our Service, although small, indicates that arthroscopy allows direct and detailed visualization of intra-articular structures, enabling immediate interventions, as recommended in the literature.

Fleiss's Kappa is derived from Cohen's Kappa. The main difference between Fleiss's Kappa statistic and Cohen's Kappa is that Fleiss's Kappa allows us to assess the degree of agreement between three or more observers/evaluators. In contrast, Cohen's Kappa allows us to analyze only two. The value of the Kappa agreement coefficient can range from  $(-p_e/1-p_e)$  to 1. The closer the value is to 1, the greater the indication that there is agreement among the evaluators, and the closer it is to zero, the greater the indication that the agreement is purely random.

Based on the results observed, arthroscopy showed the same sensitivity of 100% in identifying lesions requiring surgical intervention. However, diagnostic accuracy in indicating the best treatment depended on the information obtained from its use. This suggests that, although MRI is highly specific, its sensitivity may be compromised, indicating that the severity of some lesions may not be accurately diagnosed by this method. The results of this study corroborate the findings of previous studies, showing that the diagnostic accuracy of MRI is inferior to arthroscopy, even considering technical variations among radiologists.<sup>10</sup>

We looked at the functional results across the board after the

re related to several factors. Its retrospective biases, small sample size, and lack of randomization may interfere with the results. However, the study was conducted with good quality images obtained. We believe that the evaluation of MR and ARM results, as well as the results of diagnostic arthroscopy, depend on the personal experience of the radiologists who perform the diagnostic interpretations, and the results are directly related to the expertise of the orthopedic surgeon who performs them. We may not have documentation of low-grade complications such as skin problems or postoperative pain, as we did not find any such records. Other potential complications, though subtle and transient,

The study highlights the high specificity and sensitivity of MRI and VA for diagnosing TFCC lesions that require surgical intervention. The combination of advanced imaging technologies and minimally invasive techniques, such as arthroscopy, can significantly improve clinical outcomes by promoting more accurate diagnoses and effective treatments.

When the Quick-Dash questionnaire was applied, we observed that 18 (64.28%) patients obtained results classified as excellent, 7 (25.00%) as good, and 3 (10.72%) as satisfactory. The study revealed high specificity and sensitivity between MRI and VA examinations, with agreement in 26 (92.86%) cases.

---

**AUTHOR'S CONTRIBUTION:** Each author contributed individually and significantly to the development of this article. HLS: study design, VGJFV: data analysis and interpretation for case description, QFM: data acquisition, GJVC: intellectual content review, IFS: study design, and DET: text writing and final approval of the manuscript to be published.

---

## REFERENCES